

Stochastic Liquid Deformation Fields: An SDE Generalisation of Closed-form Continuous-time Cells for Dynamic 3D Gaussian Splatting

Mingzhao Li*, Arghya Pal*

* School of Information Technology, Monash University, Selangor, Malaysia

E-mail: {mlii0259@student, arghya.pal}@monash.edu

Abstract—Deformable 3D Gaussian Splatting (D-3DGS) reconstructs dynamic scenes by deforming a canonical set of 3D Gaussians through a deformation field of frame time. Replacing its MLP with a stack of Closed-form Continuous-time (CfC) cells—a Liquid Neural Network that solves the Liquid Time-constant ODE in closed form—gives the field continuous-time behaviour at feed-forward cost. That closed form, however, is only the deterministic limit of a noise-driven system, and drops the stochastic term usually credited for the robustness of liquid networks. We put it back: a small Gaussian perturbation is added to the time gate of every CfC cell, turning the deterministic field into a simple stochastic (SDE) one. The noise is used only during training, needs no solver, adds no parameters, and reduces exactly to the CfC when switched off—so it costs nothing at inference. On the synthetic D-NeRF scenes the stochastic field is on par with the deterministic CfC and ahead of the MLP on most scenes; on the real-world NeRF-DS scenes the deterministic limit is already best and adding noise does not help. Our contribution is therefore a clean way to read the CfC field as an SDE and an honest account of when a plain noise term helps and when it does not, rather than a new state of the art.

Index Terms—4D reconstruction, dynamic 3D Gaussian splatting, liquid neural networks, stochastic differential equations, closed-form continuous-time cells.

I. INTRODUCTION

Reconstructing a dynamic 3D scene from a single moving camera—4D reconstruction—is a foundational problem in vision and graphics. Explicit 3D Gaussian Splatting [1] achieves real-time photorealistic rendering, and Deformable 3DGS (D-3DGS) [2] lifts it to dynamic scenes by learning canonical Gaussians together with a positional-encoded MLP deformation field $F_\theta(\gamma(x), \gamma(t)) \rightarrow (\Delta x, \Delta r, \Delta s)$.

From discrete MLP to liquid field: Although t is continuous, the D-3DGS MLP couples no two timesteps in its architecture; it is fitted independently at each sampled frame, so temporal smoothness is left to optimisation. A natural fix replaces F_θ with a stack of Closed-form Continuous-time (CfC) cells [3], the analytic solution of the Liquid Time-constant (LTC) ODE [4]. Each CfC cell exposes a sigmoidal time gate that bakes a smooth, learned response to t into the loss landscape, giving the field explicit continuous-time semantics while keeping inference a feed-forward pass—no numerical solver, unlike Neural-ODE methods such as ODE-GS [5].

The discarded stochasticity: The CfC is, by construction, the *deterministic* limit of a noise-driven dynamical system: the LTC ODE from which it derives was formulated as a stochastic process, and the robustness of liquid networks to jitter and irregular sampling [6], [7] is attributed precisely to that stochastic lineage. The closed form keeps the drift and drops the diffusion. Stochastic differential equation (SDE) models [8] retain the diffusion explicitly and are reported to be more robust under corrupted supervision, but only by paying a stochastic-solver cost on every forward pass. We seek the middle point: re-introduce the diffusion term *architecturally*, inside the closed-form cell, without any solver.

This paper: We propose a *stochastic liquid deformation field*. The only path through which the elapsed-time signal τ enters a CfC cell is the affine pre-activation of its sigmoidal gate; we treat that pre-activation as an Itô process and add a small Gaussian increment $\lambda \varepsilon$ ($\varepsilon \sim \mathcal{N}(0, \mathbf{I})$, with a single noise level λ) before the sigmoid—the resulting stochastic gate is (3). This is one Euler–Maruyama step of the SDE whose drift is the original deterministic gate, so $\lambda=0$ recovers the closed-form CfC exactly, and the noise—*injected only during training*—leaves inference identical to the deterministic feed-forward pass. The numerical gains are modest; we present the method as a principled interpretation and an empirical characterisation rather than a new benchmark result.

Contributions:

- A stochastic (SDE) reading of the closed-form CfC deformation field: one Gaussian perturbation of the cell time gate (3) restores the diffusion term the closed form drops. It needs no solver, adds no parameters, runs only during training, and reduces exactly to the CfC at $\lambda=0$ —so it is free at inference and the SDE-vs-CfC comparison is exactly size-matched.
- Results on D-NeRF and NeRF-DS: the liquid field, with or without noise, is competitive with or ahead of the D-3DGS MLP on the synthetic scenes and matches it on the real-world scenes, together with a compute comparison (params, inference cost).
- An honest, calibrated noise study: a small constant noise is harmless on synthetic data but does not help—and slightly hurts—on real-world data, so on these interpo-

lation benchmarks the deterministic CfC limit is already hard to beat.

II. RELATED WORK

Dynamic 3D Gaussian splatting: Following 3DGS [1], D-3DGS [2] learns a canonical Gaussian set with an MLP deformation field; Gaussian-Flow [9], Shape of Motion [10] and FLAG-4D [11] explore explicit flow and dual-deformation variants. None treats the deformation as an explicit *stochastic* continuous-time function of t .

Continuous-time and stochastic neural networks: Neural ODEs [12] parameterise a hidden-state derivative recovered by a solver; ODE-GS [5] applies a latent neural ODE to Gaussian-splat extrapolation, paying the solver cost on every forward pass. Neural SDEs [8] replace the deterministic dynamics with an Itô integral, gaining robustness to observation noise at a still higher cost. Liquid Time-constant networks [4] combine explicit time constants with input-dependent forcing, and CfC [3] derives the analytic, deterministic solution of that ODE, replacing the solver with a sigmoidal time gate. Liquid-S4 [13] brings liquid time constants to state-space models, while subsequent work characterises the robustness of liquid networks for noisy, irregular signals [6], [7]. Our method sits between the deterministic CfC and a full Neural SDE: it restores the SDE’s diffusion term inside the closed-form cell, keeping feed-forward inference.

III. METHOD

A. Background: Liquid Deformation Field

A scene is N canonical Gaussians $\{\mathcal{G}_i = (x_i, r_i, s_i, \alpha_i, c_i)\}$. For frame time $t \in [0, 1]$ a deformation field produces per-Gaussian offsets $(\Delta x_i, \Delta r_i, \Delta s_i) = F_\theta(\gamma(\text{sg}(x_i)), \gamma(t))$, where $\gamma(\cdot)$ is a NeRF positional encoding and $\text{sg}(\cdot)$ a stop-gradient. The liquid field instantiates F_θ as a stack of D CfC cells. A single cell maps an input u , hidden state h and elapsed-time signal τ to an updated state through a shared backbone ϕ , four linear heads, and a sigmoidal time gate:

$$\begin{aligned} z &= \phi([u; h]), \\ g &= \tanh(W_g z), \quad h_{\text{cand}} = \tanh(W_h z), \\ \sigma_\tau &= \sigma(W_a z \tau + W_b z), \\ h' &= g \odot (1 - \sigma_\tau) + h_{\text{cand}} \odot \sigma_\tau. \end{aligned} \quad (1)$$

Equation (1) is the closed-form solution of the LTC ODE [3], [4]. The hidden state is reset to $\mathbf{0}$ each pass and threaded through the stack, with t supplied as the elapsed-time signal τ of every cell; a linear head reads out the offsets. The canonical Gaussian set, rasteriser, $\mathcal{L}_1$ +SSIM loss, density control and Adam schedule are inherited unchanged from D-3DGS [2]. Our contribution lives entirely inside the cell (1).

B. Stochastic (SDE) Time Gate

The LTC ODE underlying (1) is a noise-driven system; the closed form is its deterministic limit and therefore discards the diffusion. We restore it at the single point through which

τ enters the cell. Writing the gate pre-activation as a latent state $s = W_a z \tau + W_b z$, we model it as a one-step Itô process

$$ds = \underbrace{(W_a z \tau + W_b z)}_{\text{drift}} + \underbrace{\lambda dW}_{\text{diffusion}}, \quad (2)$$

whose Euler–Maruyama discretisation adds a Gaussian increment $\lambda \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \mathbf{I})$ (the constant λ absorbs $\sqrt{\Delta t}$). The stochastic cell is therefore (1) with the gate replaced by

$$\sigma_\tau = \sigma(W_a z \tau + W_b z + \lambda \varepsilon), \quad \varepsilon \sim \mathcal{N}(0, \mathbf{I}). \quad (3)$$

Three properties follow. (i) *Exact reduction.* At $\lambda=0$ the increment vanishes and (3) is identical to (1); the stochastic field is a strict generalisation of the deterministic CfC, and of the D-3DGS MLP through it. (ii) *Solver-free, deterministic inference.* The noise is sampled per forward pass during training only; at test time it is disabled, so inference is the same feed-forward pass as the deterministic CfC—no stochastic solver, unlike Neural SDEs [8]. (iii) *Regularisation on t .* Because the gate is the only route by which τ acts, perturbing it perturbs exactly the field’s response to time. In expectation the gate $\mathbb{E}_\varepsilon[\sigma_\tau]$ becomes a smoother function of τ , so (3) is *intended* to act as a learnable smoothness prior—the mechanism credited for liquid-network robustness [6]—with a strength set by λ . Whether this actually helps is an empirical question, which Sec. IV answers.

From the LTC-SDE to one gate increment: Our formulation is deliberately a *first-order, single-step* realisation of the underlying stochastic dynamics, and we make the approximation chain explicit. The full liquid system is an SDE in the hidden state $h(t)$; its closed-form drift is exactly the deterministic CfC update (1). A faithful stochastic version would integrate a hidden-state diffusion over the elapsed interval with a solver—precisely the cost we set out to avoid. We instead inject a single Euler–Maruyama increment and place it on the gate pre-activation, because that scalar is the *only* channel through which elapsed time τ enters the cell (1): perturbing it is the minimal way to make the cell’s time response stochastic without a solver and without touching the two candidate states g, h_{cand} . In code this is one line—add $\lambda \varepsilon$ to $W_a z \tau + W_b z$ before the sigmoid, drawn independently per cell and per forward pass—which is exactly (3). We do not claim a higher-order or hidden-state SDE; those are the natural stronger variants, discussed in Sec. V.

C. Implementation

We use $D=6$ cells, hidden width $W=128$, backbone depth 2 and GELU activation, with a time read-out width of 32 (D-NeRF) / 64 (NeRF-DS); the backbone width matches the deterministic liquid field. Only λ is new. The noise is drawn independently per cell and per forward pass and disabled at inference, so it adds no parameters and no test-time cost (Sec. IV-D); its training overhead is one Gaussian sample and one addition per cell. All other components follow the public D-3DGS pipeline (Sec. III-A). D-NeRF scenes train for 40k iterations and NeRF-DS for 20k on a single GPU.

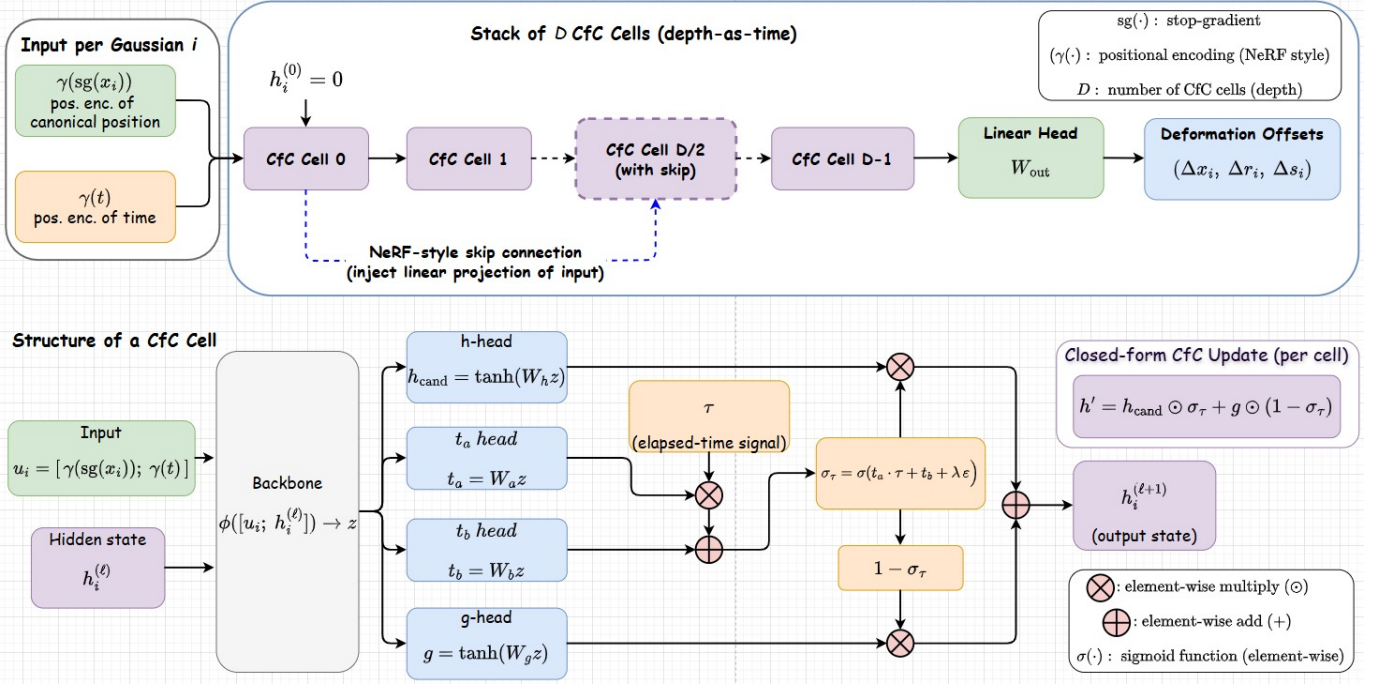


Fig. 1. **Stochastic liquid cell.** A shared backbone $\phi([u; h]) \rightarrow z$ feeds four heads: g, h_{cand} form two candidate states, while $W_a z, W_b z$ form the affine elapsed-time term. The time gate is the only path through which τ enters the cell; we add a Gaussian diffusion increment $\lambda \epsilon$ to its pre-activation ($\sigma_\tau = \sigma(W_a z \tau + W_b z + \lambda \epsilon)$), an Euler-Maruyama step of the SDE whose drift is the deterministic gate. Noise is active in training only.

IV. EXPERIMENTS

A. Setup

We evaluate on D-NeRF [14] (eight synthetic monocular scenes at 800×800) and NeRF-DS [15] (seven real-world scenes with specular, dynamic objects). We report PSNR, SSIM and LPIPS. Two baselines are retrained under an identical protocol and identical seeds: the original D-3DGS MLP field [2], and the deterministic Liquid CfC field, i.e. our model at $\lambda=0$. The proposed SDE-LNN uses the calibrated noise level $\lambda=0.05$ unless stated.

Statistical calibration: To gauge significance we retrained the identical $\lambda=0$ Hell Warrior model three times; its PSNR spans 41.48–41.57 dB ($\sigma \approx 0.05$ dB). We therefore read aggregate differences below ~ 0.1 dB as ties, and rely on (i) per-scene effects well above this floor (e.g. Hook, +1.24 dB) and (ii) the monotone λ trend of Sec. IV-E rather than single-run means.

B. D-NeRF: Eight Synthetic Scenes

Table I reports per-scene results. Against the matched D-3DGS MLP baseline, the stochastic liquid field ($\lambda=0.05$) attains the best PSNR on six of the eight scenes and a higher mean (38.28 vs. 38.19 dB), though the aggregate gap is within the variance above. Its clearest gains fall on the scenes with the most high-frequency articulated motion—Hook (+1.24 dB over the MLP), Hell Warrior (+0.38), Stand Up (+0.37) and Jumping Jacks (+0.15)—consistent with the time-jitter regularisation of the stochastic gate (3); the MLP retains a small edge on Bouncing Balls and T-Rex. This comparison

mixes architecture with capacity (Table III); the effect of the noise alone is isolated in Sec. IV-E. Fig. 2 shows the visible difference on Hell Warrior, Hook and Stand Up. All learned-deformation methods sit far above the implicit D-NeRF and TiNeuVox baselines.

C. NeRF-DS: Seven Real-World Scenes

Table II reports the seven real-world NeRF-DS scenes, which are dominated by pose noise so that the *relative* ordering of methods is the more informative quantity. The stochastic field is competitive with the matched MLP in aggregate (mean PSNR 23.73 vs. 23.82 dB) and improves the hardest manipulation scenes, taking the best PSNR on Bell and Press (+0.07 and +0.14 dB over the MLP) and the best SSIM on three scenes (Plate, Bell, Press). It trails the MLP on the cleaner Cup and As scenes, which brings the aggregate to near-parity rather than a net gain—a behaviour we examine in Sec. IV-E. Both liquid and MLP fields clearly surpass the specular-aware NeRF-DS and TiNeuVox baselines on mean PSNR.

D. Compute and Overhead

The stochastic term is free at inference. Because the noise is switched off at test time (Sec. III-B), SDE-LNN executes the *identical* feed-forward pass as the deterministic CfC, at the same throughput; it adds no parameters and, during training, only one Gaussian draw and one addition per cell. Table III lists the measured deformation-field sizes. Two points follow. First, the “does noise help” comparison (SDE-LNN vs. CfC)

TABLE I
PER-SCENE COMPARISON ON THE EIGHT D-NeRF SYNTHETIC SCENES. PSNR ($\uparrow$), SSIM ($\uparrow$) AND LPIPS ($\downarrow$) AT 800×800 TEST IMAGES. D-NeRF AND TiNeuVox NUMBERS ARE QUOTED FROM [2]; D-3DGS (MLP) AND OURS (SDE-LNN, $\lambda=0.05$) ARE RETRAINED UNDER AN IDENTICAL PROTOCOL WITH THE SAME CANONICAL-GAUSSIAN SET AND SEEDS. CELLS ARE COLOURED **BEST**, **SECOND-BEST**, **THIRD-BEST**.

Method	Hell Warrior			Mutant			Hook			Bouncing Balls		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
D-NeRF [14]	24.06	0.9440	0.0707	30.31	0.9672	0.0392	29.02	0.9595	0.0546	38.17	0.9891	0.0323
TiNeuVox [16]	27.10	0.9638	0.0768	31.87	0.9607	0.0474	30.61	0.9599	0.0592	40.23	0.9926	0.0416
D-3DGS (MLP) [2]	41.23	0.9867	0.0255	42.05	0.9944	0.0067	36.94	0.9855	0.0167	41.37	0.9957	0.0086
Ours (SDE-LNN)	41.61	0.9873	0.0259	42.08	0.9941	0.0076	38.18	0.9884	0.0144	40.22	0.9950	0.0100

Method	Lego			T-Rex			Stand Up			Jumping Jacks		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
D-NeRF [14]	25.56	0.9363	0.0821	30.61	0.9671	0.0535	33.13	0.9781	0.0355	32.70	0.9779	0.0388
TiNeuVox [16]	26.64	0.9258	0.0877	31.25	0.9666	0.0478	34.61	0.9797	0.0326	33.49	0.9771	0.0408
D-3DGS (MLP) [2]	24.92	0.9434	0.0445	37.73	0.9929	0.0103	43.85	0.9942	0.0083	37.40	0.9893	0.0141
Ours (SDE-LNN)	24.94	0.9436	0.0443	37.46	0.9926	0.0110	44.22	0.9944	0.0085	37.55	0.9896	0.0142

TABLE II
PER-SCENE COMPARISON ON THE SEVEN NeRF-DS REAL-WORLD SCENES. NeRF-DS AND TiNeuVox NUMBERS ARE QUOTED FROM [2]; D-3DGS (MLP) AND OURS (SDE-LNN, $\lambda=0.05$) ARE RETRAINED UNDER AN IDENTICAL PROTOCOL. COLOURING AS IN TABLE I.

Method	Sieve			Plate			Bell			Press		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
NeRF-DS [15]	25.78	0.8900	0.1472	20.54	0.8042	0.1996	23.19	0.8212	0.1867	25.72	0.8618	0.2047
TiNeuVox [16]	21.49	0.8265	0.3176	20.58	0.8027	0.3317	23.08	0.8242	0.2568	24.47	0.8613	0.3001
D-3DGS (MLP) [2]	25.28	0.8721	0.1514	20.50	0.8140	0.2265	25.07	0.8389	0.1637	25.35	0.8618	0.1939
Ours (SDE-LNN)	25.43	0.8690	0.1627	20.43	0.8142	0.2314	25.14	0.8399	0.1702	25.49	0.8643	0.2015

Method	Cup			As			Basin			Mean		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
NeRF-DS [15]	24.91	0.8741	0.1737	25.13	0.8778	0.1741	19.96	0.8166	0.1855	23.60	0.8494	0.1816
TiNeuVox [16]	19.71	0.8109	0.3643	21.26	0.8289	0.3967	20.66	0.8145	0.2690	21.61	0.8241	0.3195
D-3DGS (MLP) [2]	24.81	0.8893	0.1560	26.19	0.8833	0.1814	19.56	0.7956	0.1882	23.82	0.8507	0.1802
Ours (SDE-LNN)	24.08	0.8793	0.1864	25.99	0.8810	0.1901	19.55	0.7902	0.2025	23.73	0.8483	0.1921

TABLE III
DEFORMATION-FIELD PARAMETER COUNTS, MEASURED WITH PTFLOPS [17] ON THE EXACT CONFIGURATIONS USED. THE DETERMINISTIC CfC AND SDE-LNN ARE ARCHITECTURALLY IDENTICAL—THE TRAINING-TIME NOISE ADDS NO PARAMETERS AND IS DISABLED AT INFERENCE—SO THEY SHARE A ROW. THE MLP COUNT IS THE D-3DGS DEFORMATION FIELD [2].

Deformation field	Params (M)	Inference
D-3DGS MLP ($D=8, W=256$) [2]	0.52	feed-forward
Liquid CfC /SDE-LNN — NeRF-DS	0.32	feed-forward
Liquid CfC /SDE-LNN — D-NeRF	1.55	feed-forward

is exactly size-matched, so the neutral result of Sec. IV-E is not a capacity artifact. Second, the liquid field’s own size is a design choice, not fixed by the method: on NeRF-DS it is 39% smaller than the MLP, whereas the D-NeRF runs use a wider backbone. All three fields keep feed-forward inference and need no numerical solver.

E. Noise-level Study

Table IV sweeps λ and reports the dataset means, and the picture is simple: the noise never buys an aggregate gain. On D-NeRF a small λ is harmless—the mean is unchanged up to $\lambda=0.05$ —while a larger λ slowly washes out real motion and the mean falls. On NeRF-DS the deterministic limit ($\lambda=0$)

TABLE IV
NOISE-LEVEL STUDY: DATASET-MEAN METRICS AS A FUNCTION OF λ . $\lambda=0$ IS THE DETERMINISTIC CfC. **BEST** PER COLUMN.

λ	D-NeRF (8 scenes)			NeRF-DS (7 scenes)		
	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS
0.00	38.28	0.9856	0.0168	23.83	0.8482	0.1908
0.05	38.28	0.9856	0.0170	23.73	0.8483	0.1921
0.10	38.06	0.9853	0.0173	23.73	0.8480	0.1912
0.20	37.91	0.9854	0.0175	23.80	0.8489	0.1903

already gives the best PSNR, and every positive λ is a little worse. So a plain constant noise does not improve real-world reconstruction: the deterministic CfC is the sweet spot, and $\lambda=0.05$ is just the most noise one can add on synthetic data before quality starts to fall.

F. Qualitative Results

Fig. 2 compares the three D-NeRF scenes with the largest PSNR gain over the MLP. The differences are localised to the fast-moving parts: on each scene the MLP smears or leaves a faint trailing ghost around the articulated limb, while the stochastic liquid field keeps that edge sharper and closer to the ground-truth silhouette. This is the visible counterpart of the numbers in Table I.

V. DISCUSSION

Why the noise does not help here: These benchmarks supervise densely in time and ask only for interpolation *inside* the observed window—a regime the deterministic CfC already fits well, so a noise term has little error to correct, and on the noisier real-world scenes it does more harm than good. The plain constant noise is thus neutral at best: an honest negative result in which the SDE view’s value is a clean formulation and a free, tunable knob, not an automatic quality gain.

What would help: adaptive stochasticity: Noise helps only where supervision is most jittery—fast D-NeRF motion, a few hard NeRF-DS scenes—which is what it is meant to damp. A smarter noise than our constant one should do more: *annealing* λ to zero late in training, making λ *input-dependent*, or moving the diffusion from the gate to the *hidden state* (a closer match to the true LTC-SDE). All keep the solver-free, deterministic-inference property.

Robustness regime and limitations: The clean, dense D-NeRF/NeRF-DS protocol under-tests the setting the noise targets; the informative stress test is corrupted or sparse supervision (jittered timestamps, pose noise, few-view training). We leave three additions to future work: this stress test; a qualitative comparison on the real-world scenes, where the small Bell and Press wins are best shown together; and inference-time Monte-Carlo sampling for uncertainty. We study only one constant, gate-level λ ; on clean D-NeRF the method gaps sit within the run-to-run variance of Sec. IV.

VI. CONCLUSION

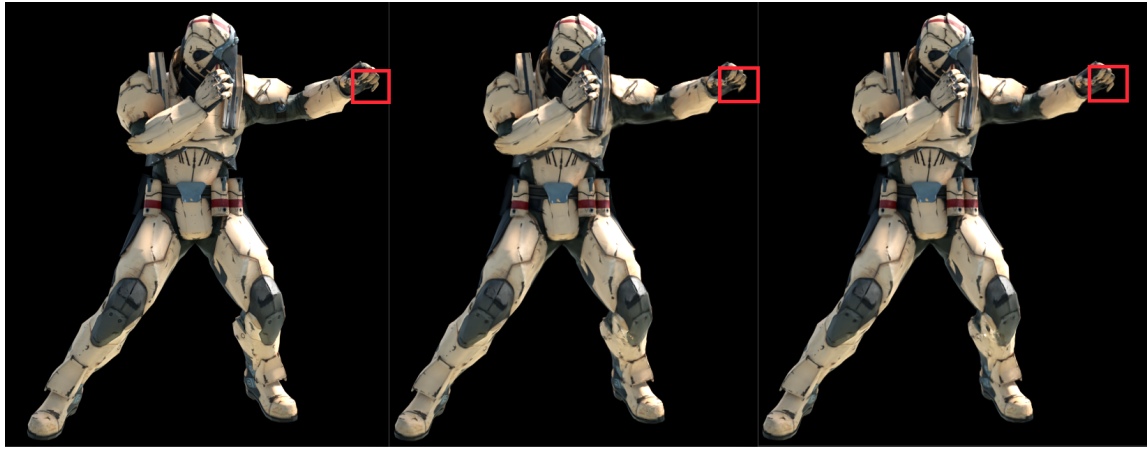
We presented a stochastic view of the closed-form CfC deformation field for Deformable 3D Gaussian Splatting: treating the cell time gate as a one-step SDE and adding a Gaussian noise to it restores the diffusion term the closed form drops, with no solver, no parameters, and deterministic inference. On D-NeRF the resulting field matches the deterministic CfC and is ahead of the MLP; on NeRF-DS the deterministic limit stays best and a plain constant noise does not help. The take-away is twofold: the CfC field has a clean SDE interpretation with a free noise knob, and, on standard interpolation benchmarks, that knob is best left near zero. Making the noise pay off on noisy, real-world supervision—through annealed, input-dependent, or hidden-state schedules—is the natural next step.

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(a) Hell Warrior (+0.38 dB): the MLP blurs the swung limb; SDE-LNN keeps its outline.



(b) Hook (+1.24 dB, the largest gain in Table I): the fast arm is ghosted by the MLP, crisp with SDE-LNN.



(c) Stand Up (+0.37 dB): the eyebrow and facial detail are blurred by the MLP but sharper with SDE-LNN.

Fig. 2. **Qualitative comparison on D-NeRF.** The three scenes with the largest PSNR gain over the MLP (Table I). Each row is one scene; within a row, left to right: **ground truth**, **D-3DGS (MLP)**, **Ours (SDE-LNN)**. Look at the fast-moving limb in each scene, where the MLP smears or ghosts the motion.